

Q1.

- (a) Find $\int x^2 e^{-x} dx$. (4)

- (b) Hence evaluate $\int_0^{\infty} x^2 e^{-x} dx$, showing the limiting process used. (3)
(Total 7 marks)

Q2.

- (a) Explain why $\int_1^{\infty} \frac{\ln x^2}{x^3} dx$ is an improper integral. (1)

- (b) (i) Show that the substitution $y = \frac{1}{x}$ transforms $\int \frac{\ln x^2}{x^3} dx$ into $\int 2y \ln y dy$. (2)

- (ii) Evaluate $\int_0^1 2y \ln y dy$, showing the limiting process used. (5)

- (iii) Hence write down the value of $\int_1^{\infty} \frac{\ln x^2}{x^3} dx$. (1)
(Total 9 marks)

Q3.

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- (a) Explain why $\int_1^{\infty} 4xe^{-4x} dx$ is an improper integral. (1)

- (b) Find $\int 4xe^{-4x} dx$. (3)

- (c) Hence evaluate $\int_1^{\infty} 4xe^{-4x} dx$, showing the limiting process used. (3)
(Total 7 marks)

Q4.

- (a) Use integration by parts to show that $\int \ln x dx = x \ln x - x + c$, where c is an arbitrary constant. (2)

- (b) Hence evaluate $\int_0^1 \ln x dx$, showing the limiting process used.

(4)

(Total 6 marks)

Q5.

- (a) Explain why $\int_1^{\infty} xe^{-3x} dx$ is an improper integral.

(1)

- (b) Find $\int xe^{-3x} dx$.

(3)

- (c) Hence evaluate $\int_1^{\infty} xe^{-3x} dx$, showing the limiting process used.

(3)

(Total 7 marks)

Q6.

- (a) Find $\int x^3 \ln x dx$.

(3)

- (b) Explain why $\int_0^e x^3 \ln x dx$ is an improper integral.

(1)

- (c) Evaluate $\int_0^e x^3 \ln x dx$, showing the limiting process used.

(3)

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(Total 7 marks)

Q7.

- (a) Explain why $\int_0^e \frac{\ln x}{\sqrt{x}} dx$ is an improper integral.

(1)

- (b) Use integration by parts to find $\int x^{-\frac{1}{2}} \ln x dx$.

(3)

- (c) Show that $\int_0^e \frac{\ln x}{\sqrt{x}} dx$ exists and find its value.

(4)

(Total 8 marks)

Q8.

- (a) Write down the value of

$$\lim_{x \rightarrow \infty} x e^{-x}$$

(1)

- (b) Use the substitution $u = x e^{-x} + 1$ to find $\int \frac{e^{-x}(1-x)}{x e^{-x} + 1} dx$.

(2)

- (c) Hence evaluate $\int_1^{\infty} \frac{1-x}{x+e^x} dx$, showing the limiting process used.

(4)

(Total 7 marks)

Q9.

- (a) Find $\int_0^a x e^{-2x} dx$, where $a > 0$.

(5)

- (b) Write down the value of $\lim_{a \rightarrow \infty} a^k e^{-2a}$, where k is a positive constant.

(1)

- (c) Hence find $\int_0^{\infty} x e^{-2x} dx$.

(2)

(Total 8 marks)

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